

eQRAP Expedice: admin guide

This guide is for the **admin** of eQRAP Expedice, the electronic QRAP V3.0 form for the warehouse and expedice. It covers:

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The app is two files: **Code.gs** (server) and **Index.html** (everything you see). The data lives in one Google Sheet. All example people use example.com addresses.

1. What the app is

- It is the firm's **QRAP V3.0 form** (logistics version), sections ① to ⑦, with the same questions and QRQC rules. Only the APU and plant-level parts, which belong to production, were left out.
- **Everything can be done on one kiosk tablet:**
 - The operator fills ①, ② and the ③ risk question, clicks **Odeslat**, and the screen returns to the board.
 - The TL finishes ③, the leader signs ④, and the pilots fill ⑤ and ⑥.
 - A **manager** closes the QRAP.
 - At every signature the kiosk asks "**Kdo podepisuje?**" and checks the name and role against the Lidé tab.
- **The status is calculated automatically** from what has been filled in. Nobody sets it by hand.
- **It is fast:** the page receives all open QRAPs when it opens, so clicking, filtering and searching need no server call. The server is called only when something is saved, plus a quiet refresh every 90 seconds.
- **It is Czech only.** The user interface and e-mails are in Czech; the code comments are in English.

2. Setting it up (one time, about 10 minutes)

1. Create a **new Google Sheet**, for example "eQRAP Expedice". Use the account that should own the app; a team account is best (see §14).
2. In the Sheet, open **Extensions** → **Apps Script**:
 - In **Code.gs**, delete everything and paste the content of **eqrap-lite/Code.gs**.
 - Click + → **HTML**, name the file **Index** (no extension), and paste **eqrap-lite/Index.html**.
 - (Optional) In **Project Settings**, tick "Show appsscript.json" and paste **eqrap-lite/appsscript.json**. If you skip this, set the time zone to **Europe/Prague** in Project Settings.
 - Click  **Save**.
3. In the toolbar, choose the function **setup**, click **Run** and allow the permissions. This creates 8 tabs and a Drive folder "eQRAP fotky".
4. Reload the Sheet. A menu **eQRAP** appears. Then:
 - **Lidé**: replace the example people with real people. You can do this later in the app on the Správa page instead.
 - **Seznamy**: check the zones, shifts and dropdown lists.
 - **Nastavení**: fill in EMAIL_KVALITA and EMAIL_BOZP. Set EMAILY to NE while you test.
5. In Apps Script, click **Deploy** → **New deployment** → **Web app** and set:
 - **Execute as**: **Me**
 - **Who has access**: **Anyone within [your company]**
 - Click **Deploy**. The URL you get is the app.
6. **Kiosk**: create one Google account just for the kiosk and add it to Lidé with the role **KIOSK**. Sign in with it on the tablet and open the URL. The app switches to kiosk mode by itself (bigger buttons, automatic return to the board, signature by name).
7. (Optional) **eQRAP** → **Vložit ukázková data** adds 3 example QRAPs. Do this in a separate test Sheet.

- "Moje konečné akce" appears above the table when you are the pilot of an open action.
- "Zobrazit i starší uzavřené" loads closed QRAPs older than ZOBRAZIT_UZAVRENE_DNI days.
- "Zrušené" link (managers only) shows cancelled QRAPs.
- The board refreshes by itself every 90 seconds. **Obnovit** refreshes now.

Nový QRAP: the form for the operator

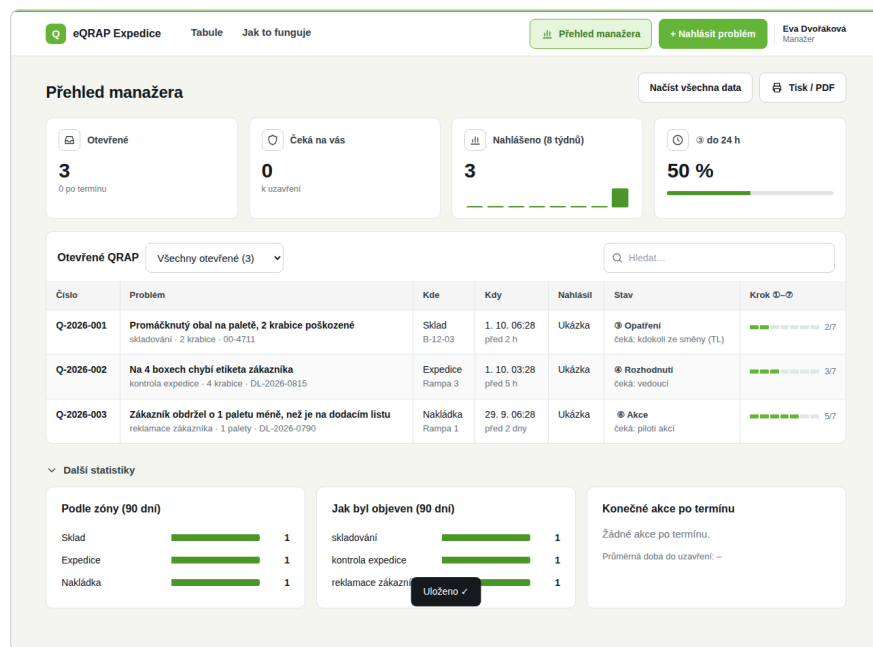
- The tabs across the top are the steps: **Problém** → **Kdy a kde** → **Množství** → **Fotky** → **Upozornění** → **Riziko** → **Kontrola**.
 - ✓ = done, ○ = not yet, green = the current step.
 - You can click a tab to go back. To go forward, use **Pokračovat** → , which checks the step first.
- **Hints:**
 - In text fields the hint is the grey placeholder; it disappears when you type.
 - Other fields have a small ⓘ next to the label. Tap it to show a one-line hint.
- **Required fields** have a red *. If something is missing, the field turns red and a message appears at the bottom.
- **Fotky:** a photo of the WRONG and the RIGHT state, or tick "Fotku nelze pořídit" and give a reason.
- **Kontrola** shows everything on one page for a last check. **Odeslat** ✓ saves the QRAP, sends the e-mails and returns to the board with a green confirmation.

QRAP detail: the form ⓘ–⑦

- **At the top:** the breadcrumb (Tabule / Q-...), the number, the status, SAFETY / ESKALACE marks, and the **Tisk / PDF** button.
- **On the right** (on a phone, at the top):

- **Další krok:** what has to happen next, who does it, what is missing, and the deadline.
- **Průběh QRAP:** ✓ for finished steps, ● for the current step, ○ for steps still ahead.
- **On the left: sections ① to ⑦** as cards.
 - The number circle is green when the section is done, solid green when it's the current section, and grey when it's waiting.
 - Buttons inside a card (Doplnit ③, Rozhodnout, Vyplnit 5 Proč, + Přidat konečnou akci, ✓ Hotovo, Podepsat uzavření, + Hodnotit...) appear **only for people who are allowed to use them**.
- **Historie změn:** click **Zobrazit** to load who changed what and when.
- **Správa QRAP (manažer),** managers only, each with a required reason: **Odemknout ①–④, Zrušit QRAP, Znovu otevřít.**
- **Tisk / PDF** prints the form without buttons and menus, or saves it as a PDF.

Přehled manažera (managers and admin)



- **Four compact cards:**
 - **Otevřené** (and how many are late)
 - **Čeká na vás** (waiting for closure)
 - **Nahlášeno** in the last 8 weeks, with a small bar strip per week. Hover a bar for the exact number.
 - **3 do 24 h:** the share of QRAPs whose immediate actions were complete within the deadline.
 - Clicking **Otevřené** or **Čeká na vás** switches the table below.
- **Otevřené QRAP:** the main table, with a switch for **Všechny otevřené / Čeká na uzavření / SAFETY** a eskalace / **Po termínu**, plus search.
- **Další statistiky** (closed until you click it):
 - QRAPs by zone and by how they were found (last 90 days). Click a bar to open the board filtered by it.
 - Overdue final actions per pilot.
 - The average time to close.
- **Načíst všechna data** also loads old closed QRAPs. **Tisk / PDF** prints the overview.

Správa (admin only)

- **Lidé a role:** add a person, edit, or deactivate. People are never deleted.
- **Seznamy ve formuláři:** one item per line (zones, how found, units, roles in ②, types of immediate actions, shifts).
- **Nastavení:** app name, e-mails on/off, Quality and BOZP e-mails, the ③ deadline, how many days of closed QRAPs to show, the kiosk timeout, and the app link for e-mails.

Changes apply immediately for you. Other users see them after reloading the page. Every change is written to the history as "SPRÁVA".

Jak to funguje

- A short help page: the five steps of the flow and a table of who can do what.
- The admin role is deliberately not shown there; it lists only Kdokoli, Vedoucí and Manažer.

4. How a QRAP moves from ① to ⑦

①②③	Operátor	– ① popis + fotky, ② kdo byl upozorněn, ③ riziko ano/ne → Odeslat
③	TL	– doplní okamžitá opatření do 24 h od KDY

- ④ Vedoucí – informované směny + rozhodnutí + podpis
 - └ „Problém vyřešen: ŽÁDNÉ 5 Proč“ → manažer schválí a uzavře
 - └ „Eskalace“ → manažeri dostanou e-mail
 - └ „Pokračovat s analýzou a 5 Proč“ → piloti
- ⑤ ⑥ Piloti – 5 Proč, kořenová příčina, konečné akce
- ⑥ TL – efektivita na 5 různých směnách
- Manažer – podepíše uzavření
- ⑦ Vedoucí / manažer – hodnocení QRQC (kdykoli, i vícekrát)

Status	Means	Who acts	Exact rule (statusOf_ in Code.gs)
③ Opatření	reported, ③ not complete	TL / anyone on the shift	s3Done is empty
④ Rozhodnutí	③ complete, waiting for the decision	VEDOUcí	no decision yet
Ke schválení	leader chose "vyřešeno bez 5 Proč"	MANAŽER	decision = VYŘEŠENO
⑤ 5 Proč	no root cause marked, or no final action yet	pilots	root0k_ false or no ⑥ action
⑥ Akce	final actions are open	pilots of the actions	some ⑥ action not done
⑥ Ověření 5 směn	all actions done; 5 different effective shifts, then closure	TL, then MANAŽER	everything else
Uzavřeno	closed by a manager	–	closedAt set
Zrušeno	cancelled by a manager (record kept)	–	cancelReason set

Important rules:

- **When ③ is complete** (the page lists anything that is missing):
 - the risk question is answered
 - if the risk is ANO, the check data (checked / wrong) is filled in and no check is still running
 - there is at least one immediate action, and all of them are done
 - "Proces zastaven?" is answered, and if ANO, the restore time is filled in
- **Signing ④** locks ①–④. Only a manager can unlock them (with a reason), and the decision must then be signed again.
- **"Vyřešeno bez 5 Proč"** is not allowed when SAFETY = ANO.
- **Each final action** removes one 5 Proč step (O1–O5 or N1–N5), or is "Aktualizovat standard".
- **Effectiveness:** 5 slots, each a **different** shift (date + shift).
 - "NEEFEKTIVNÍ" sends the QRAP back to analysis.
 - A new round of slots starts, and a new action is needed.
- **Closing checklist** (shown above the button):
 - the QRAP is in "Ověření 5 směn"
 - the root cause is marked
 - all final actions are done
 - the standard question is answered
 - 5 different effective shifts are recorded
 - "Co jsme se naučili z QR?" is filled in
 - Only a MANAŽER (or ADMIN) can sign the closure.
- **⑦ assessment** can be added many times, by leaders and managers.

5. Roles, the kiosk and e-mails

Roles (tab Lidé, column Role)

	none (operators)	VEDOUcí	MANAŽER	ADMIN
①② report + ③ risk question	✓	✓	✓	✓
③ finish immediate actions (until ④)	✓	✓	✓	✓
edit ①② of an own report before ④	own only	✓	✓	✓
④ informed shifts, sign the decision	–	✓	✓	✓
⑤ 5 Proč, ⑥ final actions	if chosen as pilot	✓	✓	✓
⑥ mark own action done, record a shift check	✓	✓	✓	✓
⑦ QRQC assessment	–	✓	✓	✓
close a QRAP, cancel / reopen / unlock, Přehled	–	–	✓	✓
Správa page	–	–	–	✓

- **Leaders and zones:** a leader with a zone in Lidé gets e-mails only for that zone. A leader with no zone gets e-mails for all zones.
- **Managers:** there can be any number, all with the same rights. If no MANAŽER exists, manager e-mails go to the ADMIN.
- **KIOSK:** only for the shared kiosk account.
- **The owner:** the owner of the script is always ADMIN, unless you use "Zobrazit jako" to test another role.

The kiosk

- Reporting, finishing ③, recording a shift check and marking an action done need **no name check**. JMÉNO in ① is typed freely.
- **Signatures** (④, ⑤ ⑥ for pilots, closing, ⑦, cancel / reopen / unlock) ask "**Kdo podepisuje?**".
 - The person types their name and picks it from the suggestions.
 - The server checks that it is an **active** person in Lidé with the **right role**. Accents and upper/lower case don't matter.
- **An ADMIN name on the kiosk** acts as a manager only. The Správa page never works on the kiosk.
- **The history** records the person's e-mail plus "(kiosk)" and the time.
- **Timeout:** after KIOSK_NAVRAT_SEKUND seconds without a touch, the kiosk returns to the board.

Who gets e-mails

When	Who
new QRAP	leaders of the zone + anyone ticked under "Poslat e-mail také" in ②
SAFETY = ANO	EMAIL_BOZP + managers
③ risk = ANO	EMAIL_KVALITA
pilots chosen in ④	the pilots
④ Eskalace	managers
new final action	the action's pilot
⑥ NEEFEKTIVNÍ	pilots + zone leaders
every morning (optional)	leaders: open QRAPS; pilots: late actions

All e-mails are switched off with EMAILY = NE .

6. Everyday admin work (no code)

I want to...	Do this
add a person, make someone a leader or manager	Správa → Lidé a role → + Přidat osobu : Jméno, e-mail of their company Google account, Role, Zóna
let someone be a pilot	add them without a role. The leader picks them in ④
someone left / take a role away	Správa → Upravit → untick Aktivní . Never delete people, because the history keeps their name
change zones, shifts or dropdown values	Správa → Seznamy : one item per line → Uložit . Old QRAPS keep their old values
set the Quality / BOZP e-mail	Správa → Nastavení
change the ③ deadline, turn e-mails off, rename the app	Správa → Nastavení
turn on the morning reminder	in the Sheet: eQRAP → Zapnout denní připomínky e-mailem
correct ①–④ after ④ was signed	as a manager, in the QRAP: Správa QRAP → Odemknout ①–④ (reason required)
remove a duplicate QRAP	Správa QRAP → Zrušit QRAP (reason required). It is hidden but kept
reopen a closed QRAP	Správa QRAP → Znovu otevřít
check what a leader / operator / kiosk sees	header Zobrazit jako (owner only), then "Admin (vy)" to return
export data	in the Sheet: File → Download → Excel , or build your own tab with formulas
back up	version history is automatic. Also do File → Make a copy once a month

7. The Google Sheet

Tab	Content	May I edit it by hand?
Lidé	Jméno, E-mail, Role, Zóna, Aktivní	✓ yes (or use Správa)
Seznamy	the dropdown values, one column per list	✓ values only; don't rename the headers
Nastavení	Klíč, Hodnota, Popis	✓ only the <i>Hodnota</i> column
Problémy	one row per QRAP; columns named after the form (① ..., ② ...)	⚠ no, use the app

Tab	Content	May I edit it by hand?
Akce	③ immediate and ⑥ final actions (column Část)	⚠ no
Ověření	the 5-shift effectiveness slots, all rounds	⚠ no
Hodnocení	⑦ assessments	⚠ no
Historie	audit trail: time, who, QRAP, what	⚠ no, never

- **Applying changes:** changes in Lidé, Seznamy or Nastavení apply within 5 minutes, or at once with **eQRAP** → **Použít změny v nastavení hned**.
- **Never:**
 - rename, delete or reorder columns in the data tabs (adding your own columns at the far right is fine)
 - delete the settings tabs
 - give users edit access to the Sheet (they don't need it; the app runs as the owner)
- **eQRAP menu** in the Sheet: Nastavit tabulku (setup) · Použít změny v nastavení hned · Zapnout / Vypnout denní připomínky · Vložit ukázková data.

8. How the code is organised

Google Sheet ↔ Code.gs (server, runs as the owner) ↔ Index.html (screen, runs in the browser)
 data + settings checks every save, sends e-mails shows data, collects input

1. **Opening the page:** doGet in Code.gs builds the page from Index.html and embeds all start data as JSON (B00T): who you are, settings, lists, people, open QRAPs, actions, checks and assessments.
2. **Saving:** the page calls a server function with google.script.run (wrapped as api('apiSave', ...) in Index.html).
3. **On the server**, every api... function:
 4. takes a lock, so two people can't write at the same time
 5. checks the permission (can..._)
 6. validates the input (...From_)
 7. writes the Sheet
 8. recalculates the status (statusOf_)
 9. writes the history (log_)
10. sends e-mails
11. returns the fresh QRAP, which the page puts into its data (store)

Golden rule: the screen only *hides* buttons for convenience. **All real checks are in Code.gs.** When you change a rule, change it in Code.gs. Change Index.html only so the screen matches.

Code.gs: the sections

§	Section	Main contents
1	Tabs, columns, defaults	TAB , COLS , SETTINGS , LISTS , EXAMPLE_PEOPLE , DECISIONS , ROLES , NEXT_STEP
2	Web app entry	doGet , startData_
3	API used by the page	apiCreate , apiSave (parts popis / opatreni / smeny / rozhodnuti / pricina / piloti / uzavreni), apiSaveAction , apiSaveEffect , apiAssess , apiSetPhoto , apiPhoto , apiHistory , apiAdmin , apiViewAs , apiAdminData , apiSavePerson , apiSaveList , apiSaveSettings
4	Status rules	statusOf_ , s3Missing_ , effState_ , checklist_ , finish_
5	Permissions	me_ , signAs_ (kiosk signatures), canPopis_ ... canAssess_ , pubQ_ (what the page receives)
6	Sheet helpers and form checks	table_ , insert_ , update_ , log_ , lock_ , cfg_ , readPeople_ ;validators popisFrom_ , opatreniFrom_ , pricinaFrom_ , defActionFields_ ; savePhoto_
7	E-mails	leaders_ , managers_ , mailNew_ , mailSafety_ , mailQuality_ , mailPilots_ , mailEscalated_ , mailAction_ , mailNotEffective_ , send_ , dailyReminder
8	Setup and menu	onOpen , onEdit , clearCache , setup , installReminder , exampleData

Index.html: the parts

Part	What it contains
<style>	the whole look. Colours, spacing and widths are variables at the top (: root)
1. State and texts	S (all data on the page), STATUS (status names), DECISIONS , CRITERIA (⑦), SECTION_TITLES

Part	What it contains
2. Small helpers	<code>h()</code> creates elements (text is always safe), <code>fill()</code> , <code>api()</code> , <code>save()</code> , <code>submit()</code> , <code>store()</code> , date formatting, <code>isLate ...</code>
3. Header and routing	<code>ICONS</code> / <code>icon()</code> , <code>go()</code> (switch page), <code>render()</code> , <code>header()</code> , <code>pageHead()</code> , <code>viewAsSelect()</code>
4. Board	<code>FILTERS</code> , <code>COLUMNS</code> , <code>boardPage()</code> , <code>rowEl()</code> (one table row), <code>myActions()</code>
5. New problem	<code>newPage()</code> : the step tabs and the <code>steps</code> list
6. Detail ①–⑦	<code>detailPage()</code> , <code>steps()</code> (progress list), <code>nextBox()</code> , <code>sec1()</code> ... <code>sec7()</code> , <code>effectBox()</code> , <code>closureBox()</code> , <code>historyBox()</code> , <code>adminBox()</code>
7. Forms	<code>makeForm()</code> (form builder), <code>popisForm</code> , <code>opatreniForm</code> , <code>decisionForm</code> , <code>pricinaForm</code> , <code>actionForm</code> , <code>assessForm</code> , <code>signerField</code>
8. Photos	<code>photosView</code> , <code>photoPicker</code> , <code>shrink</code> (resizes before upload)
9. Pages and start	<code>sparkline()</code> , <code>barList()</code> , <code>dashPage()</code> , <code>adminPage()</code> , <code>helpPage()</code> , <code>refresh()</code> , <code>startKiosk()</code> , <code>start()</code>

Tip: to find something, search (Ctrl+F) for the Czech text you see on the screen. For example, search for `'Podepsat uzavření'` to land in `closureBox`.

9. Editing `Code.gs` (server)

Rules that keep the app safe

- **Every `api...` function must check permission** with `need_(can...(me, q), 'message')` before writing anything.
- **Never trust input from the page.** Pass every value through a validator: `txt_(v, max)`, `yn_(v)`, `num_(v, label)`, `iso_(v)`, `day_(v)`, `oneOf_(v, list, msg)`, `url_(v)`.
- **Write through `insert_` / `update_`.** They put an apostrophe before text, so "00123" stays text and "=..." is never run as a formula.
- **Log every change** with `log_(id, 'what happened')`.
- **Wrap writes in `lock_()` => { ... }**.
- **Error messages** (`need_`) are shown to the user, so write them in Czech.

Columns: `COLS`

Each column is `['nameInCode', 'Header in the Sheet']`:

```
['materialNo', '① Číslo materiálu'], ['huNo', '① Číslo HU / dodacího listu'], ['supplier', '① Dodavatel'],
```

- **Order:** the order in `COLS` is the order of the columns that `setup` creates.
- **Adding:** you may add new entries anywhere. `setup` adds missing headers at the end and **never deletes data**.
- **Renaming:** don't rename a header that already has data. If you must, rename it in the Sheet *and* in `COLS` at the same time.

Settings and lists: `SETTINGS`, `LISTS`

```
['LHUTA_OPATRENI_HODIN', 24, 'Do kolika hodin od KDY mají být hotová ③ okamžitá opatření.'],
['shifts', 'Směny', ['Ranní', 'Odpolední', 'Noční']]
```

- These are only **defaults** that `setup` writes the first time. After that, the values live in the Sheet: change them on the *Správa* page, not here.
- **A new setting:** add a row here, run `setup`, then read it in code with `setting_('MY_KEY')`.
- **A new list:** add a row to `LISTS`. On the page it is available as `S.lists.myKey`.

Permissions: section 5

Each rule is one line. `me.lead = VEDOUcí/MANAŽER/ADMIN`, `me.manager = MANAŽER/ADMIN`.

```
function canDecide_(me, q) { return me.lead && q.status === 'ROZHODNUTÍ'; }
function canClose_(me, q) { return me.manager && (q.status === 'OVĚŘENÍ' || q.status === 'SCHVÁLENÍ'); }
function canAssess_(me, q) { return me.lead && q.status !== 'ZRUŠENO'; }
```

- **The page follows automatically:** `pubQ_` sends these results to the page as `q.can.decide`, `q.can.close` and so on. Buttons show or hide by themselves.
- Example: to let only managers do ⑦, change `canAssess_` to `me.manager && ...`.

Status: section 4

- `statusOf_` decides the status from the data, with the first matching rule winning.
- `s3Missing_` lists what ③ still needs.
- `checklist_` builds the closing checklist.
- If you add a status, also add it to `NEXT_STEP` in `Code.gs` and to `STATUS` (and `stage()`, `steps()`) in `Index.html`.

E-mails: section 7

Every e-mail is a list of lines:

```
function mailNew(q) {
  send_(leaders_(q.zone).concat(list_(q.notifyPeople)), q.id + ' - nový problém (' + q.zone + ')',
    ['Byl nahlášen nový QRAP.', ''].concat(facts_(q), [...]), q.id);
```

- `send_(to, subject, lines, id)` adds the app name to the subject and an **Otevřít** link at the end.
- It does nothing when `EMAILY = NE`, and a failed e-mail never stops a save.
- **To change the recipients**, change the first argument: `leaders_(zone)`, `managers_()`, `setting_('EMAIL_KVALITA')`, `list_(q.pilots)` ...

10. Editing `Index.html` (screen)

Look: colours, spacing, widths

Everything is in the variables at the top of `<style>`:

```
--ink: #151B1F; --ink-2: #374148;          /* text */
--accent: #66B539; --accent-ink: #3f7d1f; /* green buttons / green text */
--accent-soft: #E8F7DD;                  /* light green backgrounds */
--viz: #4f9a2a;                          /* chart green (darker, readable on white) */
--gutter: clamp(16px, 4vw, 72px);        /* side margins, grow with the screen */
--w-wide: 84rem; --w-mid: 72rem; --w-form: 56rem; /* max width: tables / detail / form */
--tap: 44px; --tap-s: 36px;              /* button and field height (kiosk: 52 / 44) */
```

- **Text and greens:** keep `--accent-ink` dark enough for green text on white. Text in `#66B539` is hard to read.
- **Which width a page gets** is set in `render(): { new: 'w-form', admin: 'w-mid', help: 'w-mid', detail: 'w-mid' }`. Other pages use the wide width.
- **Phone layout:** the `@media (max-width: 760px)` block.
- **Tablet layout:** the `@media (max-width: 1180px)` block.
- **Kiosk sizes:** `body.kiosk`.

Creating screen elements: `h()`

```
h('button', { class: 'btn primary', onclick: () => go('new') }, '+ Nahlásit problém')
```

- `h(tag, attributes, ...children)` builds an element.
- Text is always inserted as text, so anything a user typed can't break the page. **Don't use `innerHTML`**.
- Button styles: `btn`, plus `primary` (green), `small`, `ghost` (no border) or `danger`.

Form fields: `makeForm()`

```
const F = makeForm(values);
F.input('what', 'CO je za problém?', { req: true, rows: 3, wide: true, hint: 'Popište fakta: co je špatně a u jakého zboží.' })
F.select('how', 'JAK byl objeven?', S.lists.how, { req: true, hint: '...' })
F.yesno('safety', 'SAFETY', { req: true })
F.checks('shifts', 'Směny', S.lists.shifts)
F.radio('decision', 'Rozhodnutí', [{ value: 'A', label: '...', sub: '...' }])
F.toggle('active', 'Aktivní')
const data = F.read(); // null (and red fields + message) if something required is missing
```

Field options:

Option	Meaning
<code>req: true</code>	the field is required
<code>wide: true</code>	spans both columns
<code>rows: n</code>	makes a text area
<code>type</code>	input type: <code>'number'</code> , <code>'date'</code> , <code>'datetime-local'</code> , <code>'email'</code>
<code>now: true</code>	adds a "Ted" button
<code>list: 'people-list'</code>	name suggestions
<code>ph</code>	an explicit placeholder
<code>hint</code>	help text
<code>onchange</code>	function called when the value changes

Where hints appear:

- In a text field, `hint` becomes the placeholder.
- In any other field, it goes behind the ⓘ icon.

Show / hide: `showIf(element, condition)` . Hidden fields are never required and are saved as empty.

Board columns, statuses, filters

- **Status names and "čeká: ...":** `STATUS` in part 1.
- **Board columns:** the `COLUMNS` list in part 4 holds the header and the sort value. `rowEl()` draws the cells, in the same order.
- **Filters and count cards:** `FILTERS` . The cards shown are the `keys` list at the start of `boardPage()` .
- **Steps of the new-QRAP form:** the `steps` list in `newPage()` . Each step has a `name` (tab), an optional `title` , the content `el` and an optional `check` .

11. Recipes: common changes step by step

Before you start any recipe: **make a copy of the Sheet and test in the copy** (§12).

A. Add a new field to ⓘ (example: "Číslo zakázky")

1. **Code.gs § 1 COLS** , in the `'Problémy'` list next to `supplier` , add: `['orderNo', 'ⓘ Číslo zakázky']` ,
2. **Code.gs § 6 popisFrom_** , inside `const q = { ... }` , add: `orderNo: txt_(f.orderNo, 60)` ,
3. **Index.html part 7 popisForm** , next to the `supplier` field, add: `F.input('orderNo', 'Číslo zakázky', { hint: 'např. 4500012345' })` ,
4. **Index.html part 6 sec1** , in the `kv([...])` list, add: `['Číslo zakázky', q.orderNo]` ,
5. (Optional) to make it searchable, add `q.orderNo` to `haystack()` in part 4.
6. Run **eQRAP → Nastavit tabulku** (it adds the column), then publish a new version (§12).

B. Add a value to a dropdown (zone, shift, unit...)

No code needed: **Správa → Seznamy**, add a line, **Uložit**.

C. Add a whole new dropdown list (example: "Zákazník")

1. **Code.gs LISTS** : `['customers', 'Zákazníci', ['Zákazník A', 'Zákazník B']]` ,
2. Add a column for the value to `COLS` and to `popisFrom_` (as in recipe A). Use `oneOf_` if only listed values are allowed.
3. **Index.html popisForm** : `F.select('customer', 'Zákazník', S.lists.customers || [])` ,
4. Run setup. The list appears in **Seznamy** and on the **Správa** page.

D. Change who may do something

- Edit the one-line rule in **Code.gs § 5**. Example: only managers may assess ⓘ: `js function canAssess_(me, q) { return me.manager && q.status !== 'ZRUŠENO'; }`
- Then update the table in `helpPage()` in **Index.html**, so "Jak to funguje" stays true.

E. Change an e-mail

- **Change the text:** edit the lines in the `mailNew_` function (Code.gs § 7).
- **Add Quality to new QRAPs:** in `mailNew_` , add `.concat([setting_('EMAIL_KVALITA')])` to the recipients.

F. Change colours or sizes

- Edit the variables at the top of `<style>` (§10).
- **For the kiosk only:** edit the `body.kiosk { --tap: 52px; --tap-s: 44px; }` line.

G. Rename a status, a label or a button

- **Labels and button texts:** search for the text in **Index.html** and change the string.
- **Status names:** change the `label` in `STATUS` . Don't change the keys (`'OPATŘENÍ'` ...), because Code.gs uses them.

H. Add or reorder a column on the board

1. Add `['key', 'Header', q => sortValue]` to `COLUMNS` .
2. Add a matching `h('td', ...)` in `rowEl()` at the same position.
3. For the phone layout, give the cell a class and place it in the `@media (max-width: 760px)` block (the `.list .c-...` rules).

I. Change the 24 h deadline, the kiosk timeout or the days of closed QRAPs

No code needed: **Správa → Nastavení**.

12. Testing and publishing a change

Safe way to change the live app

1. **File** → **Make a copy** of the Sheet. The copy gets its own Apps Script project.
2. In the copy, set `EMAILY = NE`. Paste your changed code and run **setup**.
3. Open **Deploy** → **Test deployments** and try the change with **Zobrazit jako** for each role. On the kiosk, also try a signature by name.
4. When it works, paste the same code into the **live** project and run **eQRAP** → **Nastavit tabulku**.
5. Publish: **Deploy** → **Manage deployments** → **(edit)** → **Version: New version** → **Deploy**. The URL stays the same.

If you only save the code, users still see the old version. You **must** publish a new version (step 5).

Automatic tests (for developers)

The repository has a local copy of the Google services in `dev-tests/lite/`:

```
node dev-tests/lite/lite_server_test.js    # 218 checks of every rule in Code.gs
node dev-tests/lite/lite_e2e.js           # full browser run: report → close, kiosk, phone, Správa
```

- Both must end with **ALL OK**.
- The browser test also saves screenshots to `dev-tests/lite/shots/`.
- An error printed during the browser test ending with "QRAP uzavírá manažer." is expected: the test checks that a leader can't close.

13. Troubleshooting

Problem	Fix
"Aplikaci se nepodařilo spustit – chybí list / sloupec"	run eQRAP → Nastavit tabulku . Check that no header was renamed
I changed the code but nothing changed	publish a new version (§12, step 5), then reload the page
a leader has no "Rozhodnout" button	③ must be complete first (see "Další krok"). The e-mail in Lidé must match the Google account exactly, and the person must be active
kiosk: "Jméno ... není v seznamu Lidé"	the person must be active in Lidé . Pick the name from the suggestions
nobody can close a QRAP	only MANAŽER / ADMIN can. Give someone the MANAŽER role
"Podepsat uzavření" is grey	the checklist under it shows what is missing
a pilot can't edit ⑤	they must be chosen in ④, or added with "+ přidat pilota"
saving a person fails with a validation message	use the Správa page, not hand-typed values. Role must be empty or one of VEDOUCÍ / MANAŽER / ADMIN / KIOSK
I see admin rights although I set my own role lower	the owner is always ADMIN . Use Zobrazit jako to see another role
e-mails don't arrive	<code>EMAILY = ANO</code> , the address is in Lidé / Nastavení , check spam. Google sends at most about 1,500 e-mails a day
"You need permission"	the deployment must be "Anyone within [company]" and the user must be signed in with the company account
the first opening takes 2–4 s	normal: Google starts the script. After that, clicks are instant and saves take 1–2 s
a script error you don't understand	Apps Script → Executions shows every server call with the error and the line number

14. Safety and good practice

- **Ownership:** the app runs as its owner. If that account is deleted, the app and the photos stop working. Use a **team account** or a **Shared Drive**, and give a second admin edit access to the Sheet.
- **Users never need access to the Sheet or the photo folder.** Every save is checked on the server.
- **Nothing is deleted from the app:**
 - "Zrušit" keeps the record with a reason.
 - Removed actions stay marked "Odebráno".
 - Historie records everything.
- **Kiosk names are typed, not proven.** They are checked against **Lidé** and the role, and the cameras cover the rest. Give the kiosk account only the role **KIOSK**.
- **Personal data:** the app stores names and company e-mails only. Don't write health details of injuries into it.
- **Before replacing the paper QRAP,** run the app in parallel for 2–4 weeks and let Quality approve it.
- **Before editing code,** always test in a copy, keep `EMAILY = NE` while testing, and publish a new version only when it works.